

ModelArts-based ResNet50 Medical Image Classification Lab:

After-class human-in-the-loop inquiry report

Item	Submitted Information
Student name	Anonymous
Date	2026-03-19
GenAI used	DeepSeek Web Version

1. In-Class Baseline Lab Performance

Metric / Class	Baseline Result
Global accuracy	75.5%
Lordotic recall	85%
Straight recall	88%
Sigmoid recall	29%
Kyphotic recall	83%

2. Question Asked to GenAI

My prompt to DeepSeek Web Version: My ResNet50 model has 75.5% validation accuracy, but the Sigmoid class recall is only 29%. How can I improve recall for an underrepresented and difficult class?

3. GenAI Advice and Student Decision

DeepSeek Advices	Reason it may help	Student decision
Class-weighted cross-entropy	Penalizes mistakes on weak or minority classes more heavily.	Selected. It directly tests whether low Sigmoid recall is related to imbalance.
Oversampling	Shows more Sigmoid examples during training.	Not selected first. Repeating small-sample medical images may increase overfitting.
Stronger augmentation	Creates more image variation.	Not selected first. Medical X-ray augmentation must be conservative, as excessive transformations can alter the clinical meaning.

4. Revised Methodology

Original loss function

```
criterion = nn.CrossEntropyLoss()
```

Revised loss function

```
train_label_idx = train_df["Curvature"].map(label_to_idx)

class_counts = train_label_idx.value_counts().sort_index()
class_counts = torch.tensor(
    [class_counts[i] for i in range(num_classes)],
    dtype=torch.float32
)

# Inverse-frequency class balancing:
# smaller class count -> larger loss weight
class_weights = class_counts.sum() / (num_classes * class_counts)
```

```
# Additional recall pressure for Sigmoid class
# original label 3 = merged Sigmoid(3/4)
sigmoid_idx = label_to_idx[3]
class_weights[sigmoid_idx] *= 1.5

# Normalize to keep the average loss scale stable
class_weights = class_weights / class_weights.mean()
class_weights = class_weights.to(device)

criterion = nn.CrossEntropyLoss(weight=class_weights)
```

The base class weights already adjust for the training distribution. However, the baseline result showed that Sigmoid was not only slightly weaker but severely under-detected, with only 29% recall. Therefore, I added an extra multiplier to test whether stronger recall pressure could recover missed Sigmoid cases.

5. Rerun Lab Result

After retraining the model with the revised loss on Huawei Cloud ModelArts, it produced the following validation results.

Class	Precision	Recall	F1-score
Lordotic	0.79	0.68	0.73
Straight	0.76	0.55	0.64
Sigmoid	0.33	0.50	0.40
Kyphotic	0.65	0.88	0.75
Overall accuracy	61.5%		

Comparison	Baseline	After a weighted loss	Interpretation
Global accuracy	75.5%	61.5%	The overall model became worse.
Sigmoid recall	29%	50%	The model detected more Sigmoid cases.
Straight recall	88%	55%	Many Straight cases were now misclassified.

6. Error Analysis

The weighted loss did not simply improve the model; I inspected misclassified examples and Grad-CAM outputs to understand this trade-off.

Observed error pattern	What happened	Interpretation
Sigmoid predicted as Straight.	Some Sigmoid cases had Straight as top-1 and Sigmoid as top-2 with close probabilities.	The model was uncertain; the visual boundary between Straight and Sigmoid may be subtle.

Straight predicted as Sigmoid	After weighting, some Straight cases were pushed into Sigmoid.	The loss function made the model more sensitive to the Sigmoid, but also led to false positives.
Grad-CAM focus region	Heatmaps typically focus on the cervical spine.	The main issue was not irrelevant attention; the difficult part is likely visual separability.

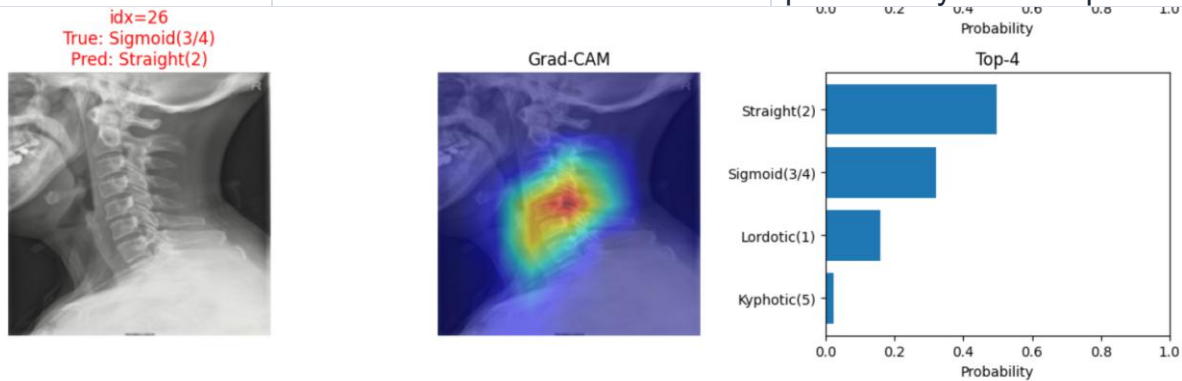


Figure 1. Example error case: true Straight image predicted as Sigmoid, with Grad-CAM focusing on the cervical spine region.

The Sigmoid weakness is not only a sample-size problem. It is also a visual-separability problem. Straight and Sigmoid X-rays can look similar, and some labels may depend on expert judgment or clinical curvature measurements rather than obvious raw image appearance.

7. Final Reflection

GenAI's advice improved the metric I targeted, but it did not produce a better final model.

First, the positive result is that Sigmoid recall increased from 29% to 50%. This means the model detected more of the originally weak class.

Second, the negative result is that overall accuracy decreased from 75.5% to 61.5%, and Straight recall decreased from 88% to 55%. This means the model over-predicted Sigmoid after the loss weighting.

My interpretation is that the X-ray image alone may not contain enough discriminative information for all borderline cases. Some Sigmoid and Straight images may differ only subtly. The class definition may also depend on anatomical landmarks, curvature angle measurement, image quality, patient posture, and expert labeling decisions. Better next steps should address data quality, visual ambiguity, and decision calibration.

Next step	Reason
Collect more Sigmoid examples.	More samples may help the model learn the internal variation of the difficult class.
Review ambiguous labels with experts.	Some Straight/Sigmoid cases may be subjective or borderline.